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Chapter 7

Robotic Sensing and Manipulation

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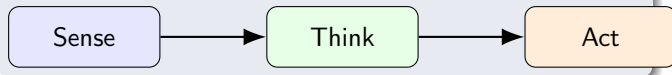
Core contents

- 1 Introduction to robotics
- 2 Sensing
- 3 Manipulation
- 4 Human-robot interaction
- 5 Navigation and path planning
- 6 Autonomous robotic systems

What is a Robot?

A **robot** is a programmable machine that can sense its environment, process information, make decisions, and perform physical actions.

Simple view



Everyday examples

- Robot vacuum cleaner maps a room and avoids obstacles.
- Drone senses height and position while flying.
- Industrial arm assembles, welds, paints or packages products.
- Medical robot assists surgeons with precise movements.

Why Robotics Matters for Computer Science

- Robotics connects **software** with the physical world.
- It uses algorithms , AI, computer vision, data science, control, networking and embedded systems.
- A robot must work under uncertainty: Computer Science helps robots become intelligent, adaptive, and autonomous.
- Robotics is a strong application area for AI



Main Types of Robots

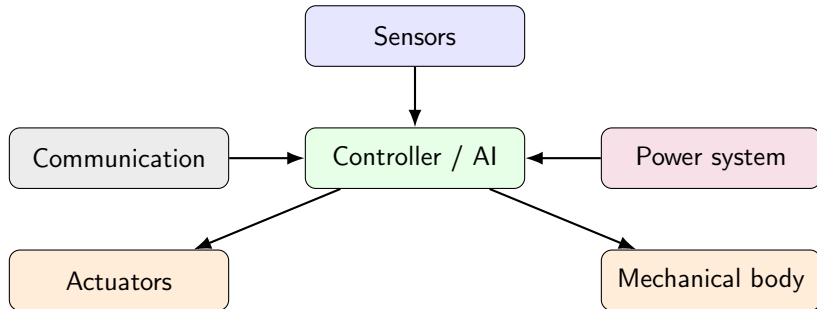
By application

- **Industrial robots:** welding, painting, assembly, packaging.
- **Service robots:** cleaning, delivery, hospitality and security.
- **Medical robots:** surgery, rehabilitation and hospital logistics.
- **Agricultural robots:** planting, spraying, harvesting and monitoring.

By mobility and body form

- **Mobile robots:** wheeled, tracked and legged robots.
- **Aerial robots:** drones and UAVs.
- **Humanoid/social robots:** human-like communication and interaction.
- **Collaborative robots:** safely work near humans.

Basic Components of a Robot



7.1.1 Robotic Sensing

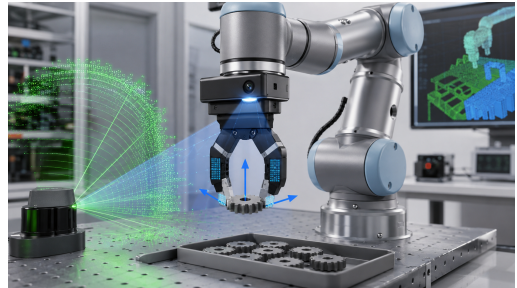
Robotic sensing means the ability of a robot to collect information from itself and from its surrounding environment.

What does sensing do?

- Measures internal state: speed, joint angle, battery and motor current.
- Measures external environment: objects, distance, light, sound and location.
- Converts physical signals into digital data for algorithms.

Common sensors

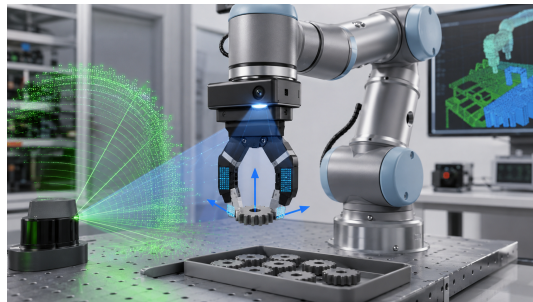
Camera, LiDAR, ultrasonic sensor, IMU, GPS, encoder, force/torque sensor, tactile sensor and microphone.



7.1.2 Manipulation

Manipulation is the ability of a robot to **physically interact** with objects.

- Reach an object using joints and links.
- Grasp the object using a gripper or hand.
- Move, place, assemble, cut, weld or sort it.
- Control position, speed and force.



Industrial Manipulation: Real-Life Example

Industrial robot arm

An industrial arm can perform repeated tasks with high speed and accuracy.

- Car manufacturing: welding and painting.
- Electronics: assembly and testing.
- Food industry: packaging and sorting.
- Warehouse: picking and palletizing.



7.1.3 Human-Robot Interaction (HRI)

HRI studies how humans and robots communicate, cooperate and share tasks safely.

- Speech and natural language interaction.
- Gesture, face and emotion recognition.
- Safety zones and collision avoidance.
- Trust, ethics and user acceptance.



Sophia-Like Social Robot: How AI is Used

AI capabilities in social robots

- Speech recognition: convert voice to text.
- Natural language processing: understand questions.
- Dialogue system: generate responses.
- Computer vision: detect faces and expressions.
- Motion control: synchronize face, eyes, mouth and hands.

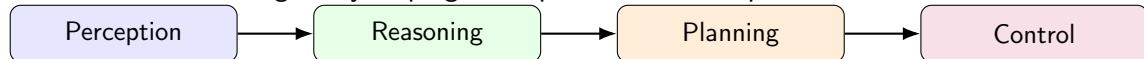
Classroom discussion

What makes a robot appear intelligent: its hardware, its software, or the interaction between both?



How AI Makes Robots Intelligent

AI makes robots intelligent by helping them perceive, reason, plan, and control their actions.



An intelligent robot can sense its environment, understand what the sensor data means, decide what action is needed, plan how to perform the action, and control its movement.

Perception	Reasoning	Planning	Control
Object detection, face recognition, SLAM, sensor fusion.	Decide what the environment means and what goal is active.	Select a path, sequence of actions or task schedule.	Move motors accurately and safely in real time.

7.2 Navigation and Path Planning

Navigation means the ability of a robot to move from one place to another safely.

Path planning means choosing the best or safest route from the starting point to the goal.

Navigation problem

A robot must answer three questions:

- ❶ Where am I?
 - ❷ Where is the goal?
 - ❸ How can I move there safely?
- Localization estimating the robot's current position..
 - Mapping builds a model of the environment.
 - Path planning chooses a route.
 - Obstacle avoidance reacts to changes.



7.2.1 Autonomous Robotic Systems

An autonomous robot can perform tasks with limited or no direct human control.

Autonomy levels

- **Manual:** human controls every action.
- **Assisted:** The robot supports the human, but the human is still the main decision-maker.
- **Supervised autonomy:** The human gives the goal, and the robot decides how to complete it.
- **Full autonomy:** robot senses, plans and acts independently.



Real-Life Examples of Autonomous Robots

Robot	Main task	AI/CS concepts used
Drone	Aerial inspection, filming, delivery, mapping	Computer vision, GPS/IMU fusion, path planning, control
Warehouse robot	Move shelves or parcels	SLAM, obstacle avoidance, task allocation
Self-driving car	Road navigation	Deep learning, perception, localization, planning
Surgical robot	Assist precise surgery	Human control, kinematics, safety and visual feedback
Social robot	Conversation and assistance	Speech recognition, NLP, face recognition, dialogue AI



Mini Case Study: Obstacle Avoiding Robot

Scenario

A small wheeled robot moves in a classroom corridor. It must detect obstacles and choose a safe direction.

- Sensor: ultrasonic or LiDAR.
- Controller: Arduino, Raspberry Pi or microcontroller.
- Actuators: DC motors.
- Algorithm: if obstacle is near, stop, turn and continue.

Pseudo-code

```
read distance
if distance < threshold:
    stop
    turn right
else:
    move forward
```

Safety

- Avoid harming humans, property and the environment.
- Use emergency stop, speed limits and safety zones.
- Test under different conditions before deployment.

Ethics

- Privacy: cameras and microphones collect sensitive data.
- Bias: AI decisions may be unfair if data is poor.
- Jobs: automation changes work roles and skills.
- Responsibility: define who is accountable when failure occurs.

- Robotics combines sensing, decision-making and physical action.
- Sensors allow robots to perceive internal and external conditions.
- Manipulation allows robots to grasp, move and interact with objects.
- HRI focuses on safe and useful communication between humans and robots.
- Navigation and path planning allow robots to move toward goals while avoiding obstacles.
- AI gives robots perception, learning, reasoning and adaptive behavior.

- R. Siegwart, I. R. Nourbakhsh and D. Scaramuzza, *Introduction to Autonomous Mobile Robots*, 2nd ed., MIT Press, 2011.
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- R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed., MIT Press, 2018.
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- Illustrative figures were generated for teaching use to represent robot types, sensing, manipulation, social interaction and navigation concepts.